Pranav Karawale

Precision *beyond* pixels.

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SUMMARY

Founding engineer specializing in high-throughput backend systems (Go, Node.js) and applied ML (Computer Vision). Scaled a consumer app to 30k+ MAU and built edge-deployed industrial IoT pipelines.

SKILLS

Languages: Python, TypeScript, Go, C, Bash
Backend: Node.js, FastAPI, PostgreSQL, MongoDB, MQTT
Infra & DevOps: AWS (EC2, S3), Docker, Traefik, Prometheus/Grafana
AI & Vision: YOLO, OpenCV, TensorFlow
IoT & Embedded: Raspberry Pi, LoRaWAN, STM32, Modbus/PLC

EXPERIENCE

Pravideon Pvt Ltd — **Jan. 2024 – Present**
Founding Engineer — Navi Mumbai, India

- Deployed a custom YOLO vision model (0.77 mAP@50) on edge hardware for autonomous fire detection; integrated with PLCs via Modbus to auto-target and control fire monitor pumps in real-time.
- Built a metadata aggregator service (Node.js, PostgreSQL) that indexed 15M+ records in under 24 hours, redesigning the pipeline to sustain continuous throughput from live data records.
- Developed a real-time IoT monitoring dashboard (Next.js, MQTT, PostgreSQL) deployed across 10+ industrial sites; achieved 99.9% uptime with automated threshold alerting for water, fire, and safety systems.
- Shipped an AWS-hosted ingestion pipeline and developed an enterprise LMS for coaching institutes that processed, tagged, and indexed 200K+ rich-media questions (text, images, LaTeX).

Avignon Université — **Jun. 2025 – Aug. 2025**
Research Intern — Avignon, France

- Developed LoRaPWN, a novel MAC-layer security enhancement for LoRaWAN, integrating ChaCha20-Poly1305 AEAD cryptography to secure IoT communications.
- Designed and benched an STM32 hardware testbed, achieving a 10x increase in cryptographic throughput and a 90% reduction in energy consumption vs standard AES implementations.
- Co-authored research detailing the LoRaPWN architecture and energy-efficiency benchmarks (manuscript in preparation for IEEE conference submission).

Freelancing — **Apr. 2023 – Dec. 2023**
AI Tooling Engineer — Remote

- Built a custom LLM orchestration framework using React JSX as a declarative prompt-structuring syntax, supporting multi-provider tool calling across OpenAI, Anthropic, and local models.
- Shipped the core abstraction layer for cross-model compatibility, enabling a client's internal AI workflow tooling.

EDUCATION

University Of Mumbai — **Nov. 2022 – Jun. 2026**
B.E. Computer Engineering

PROJECTS

Bratify Built an interactive image generator (Svelte) that organically grew to 30K monthly users; optimized rendering logic and backend caching to handle traffic spikes without downtime. [Link: https://bratify.vercel.app]

Aetherline AI digitization engine for P&ID diagrams utilizing Python, OpenCV, and YOLO. Winner of the Echelon '25 Hackathon.

AccessEye Automated ALPR and entry-management system achieving <700ms inference times on constrained edge hardware (Raspberry Pi Zero 2 W).

ACHIEVEMENTS

Winner: Echelon '25 Hackathon (AI P&ID Digitization)
Gold Medallist: JuniorSkills India 2021 (Web Technologies - National Level)
Leadership: Technical Lead, GDG on Campus VIMEET (2024-25)
OSS Contributor: Astro